

p-adic Numbers and Valuations

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Definition (1.1)

Let A be any commutative ring (unital). We define a multiplicative subset S of A as a subset that is closed under multiplication and also $1 \in S$ and $0 \notin S$ (otherwise it will be trivial).

The **localization of A by S** is a new ring RS^{-1} where elements are fractions with numerators in R and denominators in S

Localization

Definition (1.1)

Let A be any commutative ring (unital). We define a multiplicative subset S of A as a subset that is closed under multiplication and also $1 \in S$ and $0 \notin S$ (otherwise it will be trivial).

The **localization of A by S** is a new ring AS^{-1} where elements are fractions with numerators in A and denominators in S

Construction (1.2)

We construct AS^{-1} as the quotient of $A \times S$ modulo the equivalence relation

$$(a, s) \sim (b, t) \Leftrightarrow \exists u \in S \text{ such that } (at - bs)u = 0.$$

Denote the equivalence class of (a, s) as a/s . We find that AS^{-1} is a ring with additive identity $0/1$ and multiplicative identity $1/1$. Define $f(a) := a/1$; then $f : A \rightarrow AS^{-1}$ is a ring homomorphism.

For $s \in S$ we get, $f(s)^{-1} = 1/s$, as $(s/1)(1/s) = s/s = 1/1 = 1$,
thus $f(S) \subseteq (AS^{-1})^\times$.

Universal Property (1.3)

If $g : A \rightarrow B$ is a ring homomorphism with $g(S) \subseteq B^\times$, then $\exists$ a unique homomorphism h s.t $g = h \circ f$.

Proof. Define h by $h(a/s) := g(a)g(s)^{-1}$. Now, h is a ring homomorphism if it is well defined. Assume $a/s = a'/s'$, then $\exists u$ s.t. $(as' - a's)u = 0$, hence $(g(a)g(s') - g(a')g(s))g(u) = 0$. Since $g(u)$ is a unit in B , $(g(a)g(s')^{-1} = g(a')g(s)^{-1})$.

For uniqueness, if $h : AS^{-1} \rightarrow B$ is any ring homomorphism that satisfies $g = h \circ f$, then $g(a)g(s)^{-1} = h(f(a))h(f(s))^{-1} = h(f(a)f(s)^{-1}) = h((a/1)(1/s)) = h(a/s)$. Since h is determined by g , so h is unique. $\square$

When, A is an integral domain, then we always get $(a, s) \sim (b, t)$ iff $at - bs = 0$ (we can take $u = 1$ in the equivalence relation). So we can identify AS^{-1} with a subring of the fraction field of A . We also get that $f : A \rightarrow AS^{-1}$ is injective, allowing us to identify A with its image $f(A) \subseteq AS^{-1}$. So, when A is an integral domain we can view AS^{-1} as : $A \subseteq AS^{-1} \subseteq \text{Frac } A$.

Proposition (1.4).

The mappings

$$\mathfrak{q} \longmapsto \mathfrak{q}S^{-1} \text{ and } \mathfrak{p} \longmapsto \mathfrak{p} \cap A$$

are in 1 – 1-correspondences between the prime ideals $\mathfrak{q} \subseteq A \setminus S$ of A and the prime ideals $\mathfrak{p}$ of AS^{-1} .

Proof: Let $\mathfrak{q} \subseteq A \setminus S$ be a prime ideal of A , then

$$\mathfrak{p} = \mathfrak{q}S^{-1} = \left\{ \frac{q}{s} \mid q \in \mathfrak{q}, s \in S \right\}$$

is a prime ideal of AS^{-1} . Since, if for $\frac{a}{s}, \frac{a'}{s'} \in AS^{-1}$, $\frac{aa'}{ss'} \in \mathfrak{p}$, then, $\frac{aa'}{ss'} = \frac{q}{s''}$, $\implies s''aa' = qss' \in \mathfrak{q}$. Hence $aa' \in \mathfrak{q}$, because $s'' \notin \mathfrak{q}$, and so a or $a' \in \mathfrak{q}$, $\implies \frac{a}{s}$ or $\frac{a'}{s'} \in \mathfrak{p}$. Also, we get

$$\mathfrak{q} = \mathfrak{p} \cap A,$$

since $\frac{q}{s} = a \in \mathfrak{p} \cap A \implies q = as \in \mathfrak{q}$, since $\mathfrak{q}$ is a prime ideal $a \in \mathfrak{q}$ because $s \notin \mathfrak{q}$.

For converse, let $\mathfrak{p}$ be any prime ideal of AS^{-1} . Then $\mathfrak{p} \cap A$ is a prime ideal of A . For checking $\mathfrak{p}$ is prime let

$qq' \in \mathfrak{q} \implies \frac{qq'}{1} \in \mathfrak{p} \implies \frac{q}{1} \text{ or } \frac{q'}{1} \in \mathfrak{p} \implies \text{either } q \text{ or } q' \in \mathfrak{q}$. Also, $\mathfrak{q} \subseteq A \setminus S$. Because, if $\mathfrak{q}$ were to contain an $s \in S$, then we will have $1 = s \cdot \frac{1}{s} \in \mathfrak{p}$ as $\frac{1}{s} \in AS^{-1}$. Moreover, one has

$$\mathfrak{p} = \mathfrak{q}S^{-1}.$$

For if $\frac{a}{s} \in \mathfrak{p}$, then $a = \frac{a}{s} \cdot s \in \mathfrak{p} \cap A = \mathfrak{q}$, hence $\frac{a}{s} = a \frac{1}{s} \in \mathfrak{q}S^{-1}$. The mappings $\mathfrak{q} \mapsto \mathfrak{q}S^{-1}$ and $\mathfrak{p} \mapsto \mathfrak{p} \cap A$ are therefore inverses of each other, and hence proved. □

Examples : 1.) Let $\mathfrak{p}$ be a prime ideal of A . Then $S = A - \mathfrak{p}$ is multiplicatively closed (also $A - \mathfrak{p}$ is multiplicatively closed $\Leftrightarrow \mathfrak{p}$ is prime). We write $A_{\mathfrak{p}}$ instead of $S^{-1}A$ in this case. Here $\mathfrak{p}S^{-1}$ form an ideal $\mathfrak{m}$ in $A_{\mathfrak{p}}$. If $b/t \notin \mathfrak{m}$, then $b \notin \mathfrak{p}$, hence $b \in S$ and therefore b/t is a unit in $A_{\mathfrak{p}}$. Hence we get that if I is an ideal in $A_{\mathfrak{p}}$ and $I \not\subseteq \mathfrak{m}$, then I contains a unit and so its the whole ring. Hence $\mathfrak{m}$ is the only maximal ideal in $A_{\mathfrak{p}}$; hence, $A_{\mathfrak{p}}$ is a local ring. The process of passing from A to $A_{\mathfrak{p}}$ is called localization at $\mathfrak{p}$.

2). If $A = \mathbb{Z}$, $\mathfrak{p} = (p)$, p a prime number; then,

$$A_{\mathfrak{p}} = \mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, p \nmid b \right\}$$

Proposition (1.5)

Consider A_p with the unique maximal ideal, $\mathfrak{m}_p = pA_p$. Then there is a natural embedding

$$A/p \longrightarrow A_p/\mathfrak{m}_p,$$

identifying $A_p/\mathfrak{m}_p$ with the field of fractions of A/p . In particular, if p is a maximal ideal of A , then we have

$$A/p^n \cong A_p/\mathfrak{m}_p^n \quad \text{for } n \geq 1.$$

Proof: Let us consider the homomorphism

$$f : A/p^n \longrightarrow A_p/\mathfrak{m}_p^n, \quad a \bmod p^n \longmapsto a \bmod \mathfrak{m}_p^n.$$

For $n = 1$, f is injective because $p = \mathfrak{m}_p \cap A$. Now $A_p/\mathfrak{m}_p$ is a field, hence it becomes the field of fractions of A/p . Let p be maximal and $n \geq 1$. For every $s \in A \setminus p$ we have the identity $p^n + sA = A$, i.e., $\bar{s} = s \bmod p^n$ is a unit in A/p^n . For $n = 1$ this comes from the maximality of p , and for $n \geq 1$ it follows by induction:

$$A = p^{n-1} + sA \Rightarrow p = pA = p(p^{n-1} + sA) \subseteq p^n + sA \Rightarrow p^n + sA = A.$$

For injectivity of f : let $a \in A$ be s.t $a \in \mathfrak{m}_{\mathfrak{p}}^n$, i.e., $a = b/s$ with $b \in \mathfrak{p}^n$, $s \notin \mathfrak{p}$. Then $as = b \in \mathfrak{p}^n$, so that $\bar{a}\bar{s} = 0$ in $A/\mathfrak{p}^n$, and hence $\bar{a} = 0$ in $A/\mathfrak{p}^n$.

For surjectivity of f : let $a/s \in A_{\mathfrak{p}}$, $a \in A$, $s \notin \mathfrak{p}$. Then by the above formula, $\exists$ an $a' \in A$ s.t $a \equiv a's \pmod{\mathfrak{p}^n}$. Therefore $a/s \equiv a' \pmod{\mathfrak{p}^n A_{\mathfrak{p}}}$, i.e., $a/s \pmod{\mathfrak{m}_{\mathfrak{p}}^n}$ lies in the image of f . $\square$

Definition.-Discrete Valuation ring (1.6)

A discrete valuation ring is a principal ideal domain $\mathcal{O}$ with a unique maximal ideal $\mathfrak{p} \neq 0$. (localring + PID)

The maximal ideal is of the form $\mathfrak{p} = (\pi) = \pi\mathcal{O}$, for some prime element π . Since every element not contained in $\mathfrak{p}$ is a unit, it lead us that that, up to associated elements, π is the only prime element of $\mathcal{O}$. Hence, every nonzero element of $\mathcal{O}$ can be written as $\varepsilon\pi^n$, for some $\varepsilon \in \mathcal{O}^*$, and $n \geq 0$. More generally, every element $a \neq 0$ of the field of fractions K can be uniquely written as

$$a = \varepsilon\pi^n, \quad \varepsilon \in \mathcal{O}^*, \quad n \in \mathbb{Z}.$$

Directed set (2.1)

It is a set I with a relation " $\leq$ " that is reflexive ($i \leq i$), anti-symmetric ($i \leq j \leq i \Rightarrow i = j$), and transitive ($i \leq j \leq k \Rightarrow i \leq k$), in which for every i, j , there exists a k such that $i \leq k$ and $j \leq k$.

Inverse system (2.2)

Inverse system or projective system in a category is a family of rings $\{X_i : i \in I\}$ indexed by a directed set I with a family of homomorphisms $\{f_{ij} : X_j \rightarrow X_i : i \leq j\}$ such that

- i) each f_{ii} is the identity,
- ii) $f_{ik} = f_{ij} \circ f_{jk}$ for all $i \leq j \leq k$.

Definition (2.3)

Let (X_i, f_{ij}) be an inverse system in the system of rings. The **inverse limit** (or projective limit) of (X_i, f_{ij}) inverse system is defined as :

$$X := \varprojlim X_i := \left\{ x \in \prod_{i \in I} X_i : x_i = f_{ij}(x_j) \text{ for all } i \leq j \right\} \subseteq \prod_{i \in I} X_i$$

This inverse limit is a ring. The addition and multiplication is induced from the componentwise addition and multiplication in the product $\prod_{i \in I} X_i$. Also, we get a family of homomorphism

$$\pi_i : X \rightarrow X_i$$

which we get by the restrictions of the projections $\prod X_i \rightarrow X_i$ satisfying $\pi_i = f_{ij} \circ \pi_j$ for $i \leq j$.

The inverse limit $X = \varprojlim X_i$ has the universal property that if Y is another ring with the family of homomorphisms $\psi_i : Y \rightarrow X_i$ that satisfy $\psi_i = f_{ij} \circ \psi_j$ for $i \leq j$, then there is a unique homomorphism $h : Y \rightarrow X$ satisfying $\psi_i = \pi_i \circ h$.

Definition (3.1)

Fix a prime number p . A **p-adic integer** can be defined as a formal infinite series as given below

$$a_0 + a_1p + a_2p^2 + \dots$$

where $0 \leq a_i < p$, $\forall i \geq 0$. This set is denoted as $\mathbb{Z}_p$

p-adic Expansion (3.2)

Let $r \in \mathbb{Z}_{(p)} (\subset \mathbb{Q})$, i.e $r = p^k \frac{n}{d}$ where $k \geq 0$ and $(nd, p) = 1$. We expand r similarly as we do in case of decimal expansion. Here we denote the integer k as the p-adic valuation of $r = (v_p(r))$ and p^{-k} as its p-adic absolute value ($= |r|_p$). The division step consists of writing

$$r = ap^k + r'$$

where a is an integer s.t $0 \leq a < p$ and $r' = 0$ or a rational number s.t

p-adic Expansion

$|r'| < p^{-k}$ (i.e. $v_p(r') > k$). We compute this term using the bezout identity where since d and p are coprime, $\exists$ integers t and u s.t. $td + up = 1$. So

$$r = p^k \frac{n}{d} (td + up) = p^k nt + p^{k+1} \frac{un}{d}$$

Then breaking down nt by p using euclidean division gives

$$nt = qp + a$$

with $0 \leq a < p$. Hence

$$r = ap^k + p^{k+1} \frac{qd + un}{d}$$

with r' as the second term which is rational with $|r'|_p \leq p^{-k}$. So, the p-adic expansion of r is the formal power series $r = \sum_{i=k}^{\infty} a_i p^i$ which we obtained by repeating this process indefinitely.

The division step is unique because

$$p^k a_1 + p^{k+1} r_1 = p^k a_2 + p^{k+1} r_2 \implies a_1 - a_2 = p(r_1 - r_2) \implies p | (a_1 - a_2)$$

Since $|a_1 - a_2| < p \implies a_1 = a_2$. Hence, p -adic expansion of r is unique. Now we extend this expansion to any rational number ' r ' given by $r = p^{-m} \frac{g}{h}$ where $g, h \in \mathbb{Z}$, $(gh, p) = 1$. So, if

$$a_0 + a_1 p + a_2 p^2 + \dots$$

is the p -adic expansion of $\frac{g}{h} \in \mathbb{Z}_{(p)}$, then we say r has the p -adic expansion

$$a_0 p^{-m} + a_1 p^{-m+1} + \dots + a_m + a_{m+1} p + \dots$$

which we say as the p -adic number which we are going to define.

Definition (3.3)

A **p -adic Number** is the formal infinite series

$$a_{-m} p^{-m} + a_{-m+1} p^{-m+1} + \dots + a_0 + a_1 p + \dots$$

where $m \in \mathbb{Z}$ and $0 \leq a_i < p$, $\forall i \geq -m$. This set is denoted as $\mathbb{Q}_{(p)}$

Now here we get a natural mapping from $\mathbb{Q} \hookrightarrow \mathbb{Q}_{(p)}$ which is injective as the expansion is unique. So it also takes $\mathbb{Z} \hookrightarrow \mathbb{Z}_{(p)}$. Thus for every rational number r we get an identity,

$$r = \sum_{i=-m}^{\infty} a_i p^i$$

Different view of $\mathbb{Z}_p$ Let

$$s_n = \sum_{i=0}^{n-1} a_i p^i \in \mathbb{Z}$$

. Now define the sequences of the residue classes as

$$\bar{s}_n \equiv s_n \pmod{p^n} \in \mathbb{Z}/p^n\mathbb{Z}$$

We also observe that

$$\bar{s}_{n+1} \equiv \bar{s}_n \pmod{p^n} \in \mathbb{Z}/p^n\mathbb{Z}$$

This are natural surjective homomorphism given by

$$\mathbb{Z}/p\mathbb{Z} \xleftarrow{\lambda_1} \mathbb{Z}/p^2\mathbb{Z} \xleftarrow{\lambda_2} \mathbb{Z}/p^3\mathbb{Z} \xleftarrow{\lambda_3} \dots$$

where

$$\lambda_n(\bar{s}_{n+1}) = \bar{s}_n$$

This forms the **inverse limit** of the rings $\mathbb{Z}/p^n\mathbb{Z}$, denoted by $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ ($= \tilde{\mathbb{Z}}_p$ say). So we get,

$$\tilde{\mathbb{Z}}_p = \varprojlim \mathbb{Z}/p^n\mathbb{Z} = \{(x_n)_{n \in \mathbb{N}} \in \prod_{n=1}^{\infty} \mathbb{Z}/p^n\mathbb{Z} \mid \lambda_n(x_{n+1}) = x_n, n = 1, 2, 3, \dots\}$$

Theorem (3.4)

$$\mathbb{Z}_p \cong \tilde{\mathbb{Z}}_p. (\text{bijection})$$

Proof. Consider the function

$$\begin{aligned} \Phi : \mathbb{Z}_p &\longrightarrow \tilde{\mathbb{Z}}_p, \\ a &\longmapsto (\phi_1(a), \phi_2(a), \dots) \end{aligned}$$

where for $i \geq 1$ and $a = \sum_{k=0}^{\infty} a_k p^k$ (it's p -adic expansion), we have

$$\phi_i : \mathbb{Z}_p \longrightarrow \mathbb{Z}/p^i\mathbb{Z}$$

$$a \longmapsto \sum_{k=0}^{i-1} a_k p^k \pmod{p^i}.$$

Now, choose $(a_1, a_2, \dots) \in \tilde{\mathbb{Z}}_p$, and for $n \geq 1$, let $0 \leq s_n \leq p^n - 1$ be the unique integer such that $s_n \equiv a_n \pmod{p^n}$. In base p , write

$$s_n = \lambda_{n,0} + \lambda_{n,1}p + \dots + \lambda_{n,n-1}p^{n-1}.$$

For $0 \leq i \leq n-1$, we have $\lambda_{n+1,i} = \lambda_{n,i} \Rightarrow \lambda_{n,i} \rightarrow \lambda_i$ (constant sequence) for some λ_i , as $n \rightarrow \infty$. Using this, let

$$\Psi : \tilde{\mathbb{Z}}_p \longrightarrow \mathbb{Z}_p$$

$$(a_1, a_2, \dots) \longmapsto \sum_{k=0}^{\infty} \lambda_k p^k.$$

We also get that $\Phi \circ \Psi$ and $\Psi \circ \Psi$ provides identity maps on $\tilde{\mathbb{Z}}_p$ and $\mathbb{Z}_p$, hence proving the theorem.

(3.5) Lemma.

$$\frac{1}{1+kp} = 1 - kp + kp^2 + \cdots \text{ where } k \in \mathbb{Z}_p$$

Proof. Let $s_n = 1 - kp + (kp)^2 - (kp)^3 + \cdots (kp)^{n-1} \in \mathbb{Z}_p$. Then we have the identity,

$$\begin{aligned} 1 &= s_n(1 + kp) + (kp)^n(-1)^n \\ \Rightarrow s_n(1 + kp) - 1 &= (kp)^n(-1)^{n+1} \\ \Rightarrow |s_n(1 + kp) - 1|_p &< \frac{1}{p^n} \leq \frac{1}{p^N} \quad \forall n \geq N \\ \Rightarrow \lim_{n \rightarrow \infty} s_n(1 + kp) &= 1 \quad (\text{wrt } |\cdot|_p) \\ \Rightarrow (1 + kp) \lim_{n \rightarrow \infty} s_n &= 1 \end{aligned}$$

Now $s = \lim_{n \rightarrow \infty} s_n = 1 - kp + (kp)^2 + \cdots \in \mathbb{Z}_p$. Hence,

$$(1 + kp)^{-1} = 1 - kp + (kp)^2 - (kp)^3 + \cdots$$

(3.6) Proposition.

The unit group $\mathbb{Z}_p^\times$ in the ring $\mathbb{Z}_p$ consists of the p -adic integers

$$\mathbb{Z}_p^\times = \left\{ \sum_{i \geq 0} a_i p^i : a_0 \neq 0 \right\}$$

Proof. Take any $x \in \mathbb{Z}_p^\times$. Since $\mathbb{Z}_p \cong \tilde{\mathbb{Z}}_p$, we can say,

$x = (s_0, s_1, s_2, \dots, s_n, \dots)$ Let, $y = x^{-1} = (t_0, t_1, t_2, \dots, t_n, \dots)$ as $y \in \mathbb{Z}_p^\times$

Then,

$$yx = (s_0 t_0, s_1 t_1, s_2 t_2, \dots) = (1, \dots, 1) \Rightarrow s_0 t_0 \equiv 1 \pmod{p}$$

Now $s_0 = a_0$. So, $a_0 t_0 \equiv 1 \pmod{p} \Rightarrow a_0 \neq 0$. This proves the inclusion

$\mathbb{Z}_p^\times \subset \left\{ \sum_{i \geq 0} a_i p^i : a_0 \neq 0 \right\}$. Conversely, choose $0 < b_0 < p$ with

$a_0 b_0 \equiv 1 \pmod{p}$ and write $a_0 b_0 = 1 + kp$. Hence, if we write

$a = a_0 + p\alpha$, where α is also a p -adic integer, then

$$a \cdot b_0 = 1 + kp + p\alpha b_0 = 1 + p\kappa$$

for some p -adic integer κ . It is enough to show that the p -adic integer $1 + \kappa p$ is unit, since then we can write

$$a \cdot b_0(1 + \kappa p)^{-1} = 1 \implies a^{-1} = b_0(1 + \kappa p)^{-1}.$$

Now, notice that from last lemma, we can take

$$(1 + \kappa p)^{-1} = 1 - \kappa p + (\kappa p)^2 - \dots = 1 + c_1 p + c_2 p^2 + \dots$$

with integers $c_i \in \{0, 1, \dots, p-1\}$. Hence proved. $\square$

Corollary (3.7)

Any p -adic integer can be written in a unique way as $a = up^k \in \mathbb{Z}_p$ where $u \in \mathbb{Z}_p^\times$ and $k \geq 0$

Proposition (3.8)

$\mathbb{Q}_p$ is the field of fractions of $\mathbb{Z}_p$

Proof. Clearly for any $f \in \mathbb{Q}_p$ we can write ,

$$f = p^{-m}g \quad \text{where } g \in \mathbb{Z}_p$$

Hence, $\mathbb{Q}_p \subseteq f.f(\mathbb{Z}_p)$. Now,

$$f.f(\mathbb{Z}_p) = \left\{ \frac{r}{s} \mid r, s \in \mathbb{Z}_p, s \neq 0 \right\}$$

. By the last corollary we can say $s = up^k$ with u as unit and $k \geq 0$. In particular,

$$\frac{r}{s} = \frac{r}{up^k} = \frac{u^{-1}r}{u^{-1}up^k} = \frac{u^{-1}r}{p^k}$$

Now $u^{-1}r \in \mathbb{Z}_p$ and hence proved. □

Definition (4.1)

If $0 \neq x \in \mathbb{Z}$, we define the p -adic valuation (or exponential valuation) of x as

$$v_p(x) = \max \{r : p^r \mid x\} \geq 0.$$

For $a/b \in \mathbb{Q}$, the p -adic valuation of a/b

$$v_p : \mathbb{Q} \longrightarrow \mathbb{Z} \cup \{\infty\}$$

$$v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b).$$

We put the convention that $v_p(0) = \infty$.

Proposition (4.2)

If $x, y \in \mathbb{Q}$, then v_p has the following properties:

- (a) $v_p(x) = \infty$ if and only if $x = 0$;
- (b) $v_p(xy) = v_p(x) + v_p(y)$;
- (c) $v_p(x + y) \geq \min \{v_p(x), v_p(y)\}$ with equality if $v_p(x) \neq v_p(y)$.

Proof. Clearly (a) and (b) are true, we will only prove (c). Let x, y be non-zero rational numbers. Write $x = p^r \frac{a}{b}$ and $y = p^s \frac{c}{d}$, where $a, b, c, d \in \mathbb{Z}$ with $p \nmid a, b, c, d$ and $r, s \in \mathbb{Z}$. Now if $r = s$, we have

$$\begin{aligned} x + y &= p^r \left(\frac{a}{b} + \frac{c}{d} \right) \\ &= p^r \frac{(ad + bc)}{bd} \end{aligned}$$

which gives $v_p(x + y) \geq r$ since $p \nmid bd$. Now suppose that $r \neq s$, say $s > r$. Then

$$\begin{aligned} x + y &= p^r \left(\frac{a}{b} + p^{s-r} \frac{c}{d} \right) \\ &= p^r \frac{(ad + p^{s-r} bc)}{bd} \end{aligned}$$

Observe that as $s - r > 0$ and $p \nmid ad$, then

$$v_p(x + y) = r = \min \{v_p(x), v_p(y)\}$$

For the case when at least one of the terms is 0, we can clearly see that $v_p(x + y) = \min \{v_p(x), v_p(y)\}$

Definition (4.3)

For $x \in \mathbb{Q}$, define the **p -adic absolute value** (multiplicative valuation) of x as

$$|x|_p = \begin{cases} p^{-v_p(x)} & \text{if } x \neq 0, \\ p^{-\infty} = 0 & \text{if } x = 0 \end{cases}$$

Proposition (4.4)

The function $|\cdot|_p : \mathbb{Q} \longrightarrow \mathbb{R}^+$ has the properties

- (a) $|x|_p \geq 0$ and 0 if and only if $x = 0$;
- (b) $|xy|_p = |x|_p |y|_p$;
- (c) $|x + y|_p \leq \max\{|x|_p, |y|_p\}$, with equality if $|x|_p \neq |y|_p$.
- (d) $|x + y|_p \leq |x|_p + |y|_p$

Hence it satisfies the conditions of norm on $\mathbb{Q}$

Proof. This follows from the last Proposition.

The usual absolute value $|\cdot|$ on $\mathbb{Q}$ will be denoted here after by $|\cdot|_\infty$ in this context.

Definition (4.5)

A **Cauchy sequence** with respect to $|\cdot|_p$ in $\mathbb{Q}$ is defined by a sequence $\{x_n\}$ of rational numbers such that for every $\varepsilon > 0$, there exists a positive integer n_0 s.t

$$|x_n - x_m|_p < \varepsilon \text{ for all } n, m \geq n_0.$$

Proposition. (4.6)

Every formal infinite series

$$\sum_{\nu=0}^{\infty} a_{\nu} p^{\nu}, \quad 0 \leq a_{\nu} < p,$$

gives us a Cauchy sequence via its partial sums

$$x_n = \sum_{\nu=0}^{n-1} a_{\nu} p^{\nu},$$

Proof. For any $n, m \geq N$ we have,

$$|x_n - x_m|_p = \left| \sum_{i=m}^{n-1} a_i p^i \right|_p \leq \max_{m \leq i < n} \{ |a_i p^i|_p \} \leq \frac{1}{p^m} \leq \frac{1}{p^N}.$$

Definition (4.7)

A sequence $\{x_n\}$ in $\mathbb{Q}$ is called a **nullsequence** with respect to $|\cdot|_p$ if x_n converges to 0 w.r.t $|\cdot|_p$ norm .(i.e $|x_n|_p$ sequence converging to 0 in the usual topology of $\mathbb{Q}$.)

Proposition (4.8)

Define the set of all Cauchy sequences of elements of $\mathbb{Q}$ as:

$$\mathcal{C} = \mathcal{C}_p(Q) = \{(x_n) : (x_n) \text{ is a Cauchy sequence with respect to } |\cdot|_p\}$$

Define the ring operations as given

$$(x_n) + (y_n) = (x_n + y_n)$$

$$(x_n) \cdot (y_n) = (x_n y_n)$$

Then $\mathcal{C}$ is a commutative ring with unity.

Proof. The additions and multiplications are commutative, associative and distributive, as it follows from $\mathbb{Q}$ which is a ring. The identity is the constant sequence $x_n = 1$. We just need to show that the sum and product of two Cauchy sequences is also a Cauchy sequence. Here,

$$|(x_n + y_n) - (x_m + y_m)|_p \leq |x_n - x_m|_p + |y_n - y_m|_p \leq 2\epsilon \quad \forall m, n \geq N$$

Also a Cauchy sequence is bounded, so we can get a bound of x_n and y_n by some large positive constants say A and B , respectively. Hence

$$\begin{aligned} |x_n y_n - x_m y_m| &= |x_n y_m - x_m y_n + x_m y_n - x_m y_m| \\ &= |y_n (x_n - x_m) + x_m (y_n - y_m)|. \end{aligned}$$

$$|x_n y_n - x_m y_m| \leq \left| B \left(\frac{\epsilon}{2B} \right) + A \left(\frac{\epsilon}{2A} \right) \right| = \epsilon \quad \forall n, m \geq M$$

completes the proof. □

We also observe that $\mathbb{Q} \subseteq \mathcal{C}$ by considering the map from some rational number x to a corresponding constant sequence (x) in $\mathcal{C}$.

Proposition (4.9)

We define $\mathcal{N} \subset \mathcal{C}$ to be the set of nullsequences,

$$\mathcal{N} = \left\{ (x_n) : x_n \xrightarrow{|\cdot|_p} 0 \right\} = \left\{ (x_n) : \lim_{n \rightarrow \infty} |x_n|_p = 0 \right\}$$

and this set is a maximal ideal of the ring $\mathcal{C}$.

Proof. $\mathcal{N} \subset \mathcal{C}$ as any converging sequence is a Cauchy sequence and it is an ideal as for any $(x_n), (y_n) \in \mathcal{N}$ and $(r_n) \in \mathcal{C}$ since $(x_n) + (y_n) = (x_n + y_n)$ and $(r_n) \cdot (x_n) = (r_n x_n)$ and also (r_n) is bounded, we get

$$x_n + y_n \xrightarrow{|\cdot|_p} 0$$
$$|r_n x_n|_p \leq B\epsilon \quad \text{for some bound } B > 0, \forall n \geq N$$

We have proved it is an ideal. Now we will prove its maximality. Let $e = (1, 1, 1, \dots)$.

Then e is the multiplicative identity of $\mathcal{C}$. Suppose $x \in \mathcal{C}, x \notin \mathcal{N}$.

To prove that $\mathcal{N}$ is a maximal ideal of $\mathcal{C}$, it is enough to show

$$(\mathcal{N}, x) = (e).$$

We know for any Cauchy sequence with norm it is convergent iff it has a convergent subsequence. Now if x has a convergent null subsequence then x will be a null sequence. Since $x \notin \mathcal{N}$, at most finitely many terms of x are zero.

So, suppose $x_n \neq 0$ for $n > N$.

Let f be the element of $\mathcal{C}$ such that $f_k = 1$ for $0 \leq k \leq N$ and $f_k = 0$ for $k > N$. Clearly $f \in \mathcal{N}$.

Let $y = x - xf + f \implies y = f + x(1 - f) \implies y \in (\mathcal{N}, x)$

Also $y \notin \mathcal{N}$ as x doesn't converge to 0. By choice of f , all terms of y are nonzero which can be given by

$$y = (1, 1, 1, \dots, 1, x_n, x_{n+1}, \dots)$$

Let $1/y$ denote the sequence $(1/y_0, 1/y_1, 1/y_2, \dots)$.

Since y is a Cauchy sequence with all terms nonzero, and $y \notin \mathcal{N}$ implying no null subsequence of y , the terms of y are bounded away from zero (i.e. $|y_n| \geq B \ \forall n \geq N$), hence the products $|y_m y_n|_p \geq B^2 \ \forall n \geq N$. Hence,

$$\left\| \frac{1}{y_m} - \frac{1}{y_n} \right\|_p = \left\| \frac{y_n - y_m}{y_m y_n} \right\|_p \leq \frac{\epsilon}{B^2} \quad \forall n, m \geq N$$

So we get that $1/y$ is a cauchy sequence that is $1/y \in \mathcal{C}$.

Now $y \in (\mathcal{N}, x)$ and $1/y \in \mathcal{C}$ that implies,

$$y(1/y) \in (\mathcal{N}, x) \implies (\mathcal{N}, x) = (e)$$

Definition (4.10)

Define the field of p -adic numbers to be the quotient of the ring $\mathcal{C}$ by its maximal ideal $\mathcal{N}$:

$$\mathbb{Q}_p = \mathcal{C}/\mathcal{N}$$

Here we have constructed a new ring $\mathbb{Q}_p$, based upon the p -adic norm $|\cdot|_p$ which we call as the **Completion** of $\mathbb{Q}$ (also denoted as $\hat{\mathbb{Q}}$). Here $\mathbb{Q}$ can be identified with the constant sequence representative of $\mathbb{Q}_p$. So $\mathbb{Q} \subset \mathbb{Q}_p$

Definition (4.10)

If λ is an element of $\mathbb{Q}_p$ and (x_n) is any Cauchy sequence in $\mathbb{Q}$ representing λ , we define

$$|\lambda|_p = \lim_{n \rightarrow \infty} |x_n|_p.$$

This limit exist as $\{|x_n|_p\}$ is a cauchy sequence in $\mathbb{R}$, since from triangle inequality, we get

$$||x_n|_p - |x_m|_p| \leq |x_n - x_m|_p \leq \epsilon \quad \forall n \geq N$$

$| \cdot |_p$ is also well-defined which we can verify easily using last inequality.

Lemma (4.11)

If x is an element of $\mathbb{Q}_p$ and (x_n) is any Cauchy sequence in $\mathbb{Q}$ representing x , then

$$x = \lim_{n \rightarrow \infty} x_n \quad \text{w.r.t } | \cdot |_p.$$

Proof. We know (x_n) is a cauchy sequence in $\mathbb{Q}$ w.r.t $| \cdot |_p$. Hence,

$$\begin{aligned} & |x_n - x_m|_p < \epsilon \quad \forall n, m \geq N \\ \Rightarrow & \lim_{m \rightarrow \infty} |x_n - x_m|_p < \epsilon \quad \forall n \geq N \\ \Rightarrow & |(x_n) - x|_p < \epsilon \quad \forall n \geq N \\ \Rightarrow & \lim_{n \rightarrow \infty} x_n = x \end{aligned}$$

Third step happens as $\{x - (x_n)\}_m = \{x_m - x_n\}_m$ for a fixed x_n .

Lemma (4.12)

. Let $(x_n) \in \mathcal{C} \setminus \mathcal{N}$. The sequence of rational numbers $(|x_n|_p)$ is eventually constant, i.e., $\exists N$ such that $|x_n|_p = |x_m|_p$ whenever $m, n \geq N$.

Proof. Since (x_n) is a Cauchy sequence which does not tend to 0, that implies it has no convergent subsequence to 0 and only finite 0's are there in sequence (from proof of (4.9)). So, we can find c and N_1 such that

$$n \geq N_1 \implies |x_n|_p \geq c > 0.$$

Also $\exists N_2$ s.t

$$n, m \geq N_2 \implies |x_n - x_m|_p < c.$$

Let $N = \max\{N_1, N_2\}$. Then by norm inequality,

$$n, m \geq N \implies |x_n - x_m|_p < \max\{|x_n|_p, |x_m|_p\}$$

and so $|x_n|_p = |x_m|_p$.



Now, if $\{x_n\}$ is the representative of the Cauchy sequence for $x \in \mathbb{Q}_p$ where $x_n \neq 0$, then

$$v_p(x_n) = -\log_p |x_n|_p$$

either diverges to ∞ or is a Cauchy sequence in $\mathbb{Z}$ which eventually becomes constant for large n by (4.11) Lemma. Thus we can extend p -adic exponential valuation v_p on $\mathbb{Q}$ to an exponential valuation.

Definition (4.13)

Define **p -adic exponential** by

$$v_p : \mathbb{Q}_p \longrightarrow \mathbb{Z} \cup \{\infty\}$$

where

$$v_p(x) = \lim_{n \rightarrow \infty} v_p(x_n) = v_p(x_{n_0}) \quad \text{for } n \geq n_0$$

We also find that for all $x \in \mathbb{Q}_p$,

$$|x|_p = p^{-v_p(x)}.$$

Proposition (4.14)

Define the set

$$\mathbb{Z}_p := \left\{ x \in \mathbb{Q}_p \mid |x|_p \leq 1 \right\}$$

Then $\mathbb{Z}_p$ is a subring of $\mathbb{Q}_p$ with the group of units,

$$\mathbb{Z}_p^* = \left\{ x \in \mathbb{Z}_p \mid |x|_p = 1 \right\}.$$

Proof: $\mathbb{Z}_p$ is closed under addition and multiplication since for any $x, y \in \mathbb{Z}_p$

$$|x + y|_p \leq \max\{|x|_p, |y|_p\} \leq 1 \text{ and } |xy|_p = |x|_p |y|_p \leq 1.$$

For any unit $x \in \mathbb{Z}_p$, let y be the inverse. Now,

$$|xy|_p = 1 \implies |y|_p \geq \frac{1}{|x|_p} \geq 1$$

But $y \in \mathbb{Z}_p$ and so $|y|_p \leq 1$. Hence, $|x|_p = |y|_p = 1$. And so,

$$\mathbb{Z}_p^* = \left\{ x \in \mathbb{Z}_p \mid |x|_p = 1 \right\}.$$

Corollary (4.15)

Every element $x \in \mathbb{Q}_p^*$ (resp. $\mathbb{Z}_p$) admits a unique representation

$$x = p^m u \quad \text{with } m \in \mathbb{Z} \text{ (resp. } \in \{\mathbb{N} \cup (0)\}) \text{ and } u \in \mathbb{Z}_p^*.$$

Proof. If $v_p(x) = m \in \mathbb{Z}$, then $v_p(xp^{-m}) = 0$, hence $|xp^{-m}|_p = 1$, i.e., $u = xp^{-m} \in \mathbb{Z}_p^*$. For $x \in \mathbb{Z}_p, |x|_p \leq 1 \implies v_p(x) = m \geq 0$. Hence same calculations follows.

Proposition (4.16)

$\mathbb{Z}_p$ is the closure of the ring $\mathbb{Z}$ w.r.t $|\cdot|_p$ norm in the field $\mathbb{Q}_p$.

Proof. If $\{x_n\}$ is a convergent sequence (also Cauchy sequence) in $\mathbb{Z}$ then by (4.11), for some $x \in \mathbb{Q}_p$ $x = \lim_{n \rightarrow \infty} x_n$; then $|x_n|_p \leq 1 \implies |x|_p \leq 1$, hence $x \in \mathbb{Z}_p$. For Converse, let $x = \lim_{n \rightarrow \infty} x_n \in \mathbb{Z}_p$, for a Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$. We observe that $|x|_p = |x_n|_p \leq 1$ for $n \geq n_0$ from (4.12). So, $x_n = \frac{a_n}{b_n}$, with $a_n, b_n \in \mathbb{Z}$, $(b_n, p) = 1 \forall n \geq n_0$. Choosing for each $n \geq n_0$ a solution $y_n \in \mathbb{Z}$ of the congruence $b_n y_n \equiv a_n \pmod{p^n}$ gives us $|x_n - y_n|_p \leq \frac{1}{p^n}$ and thus $x = \lim_{n \rightarrow \infty} y_n, \implies x$ belongs to the closure of $\mathbb{Z}$.

Proposition (4.17)

The nonzero ideals of the ring $\mathbb{Z}_p$ are the principal ideals

$$(p^n) = p^n \mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid v_p(x) \geq n\},$$

with $n \geq 0$. Hence with only one unique maximal ideal (p) , $\mathbb{Z}_p$ is a local ring and so Discrete valuation ring. Also one has

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}$$

Proof: Let $I \neq (0)$ be an ideal of $\mathbb{Z}_p$ and $x = p^m u$, $u \in \mathbb{Z}_p^*$, an element of I with smallest possible m (since $|x|_p \leq 1$, one has $m \geq 0$). For any $y = p^n u' \in I$, $u' \in \mathbb{Z}_p^*$, implies $n \geq m$, hence $y = (p^{n-m} u') p^m \in p^m \mathbb{Z}_p$ and for converse $(x) = p^m \mathbb{Z}_p \subseteq I$. Then $I = p^m \mathbb{Z}_p$. Now, the homomorphism

$$\mathbb{Z} \longrightarrow \mathbb{Z}_p / p^n \mathbb{Z}_p, \quad a \longmapsto a \bmod p^n \mathbb{Z}_p,$$

has kernel $p^n \mathbb{Z}$ as any element will have valuation of $\geq n$. It is also surjective.

Since, for every $x \in \mathbb{Z}_p$, there exists by (4.16) an $a \in \mathbb{Z}$ s.t

$$|x - a|_p \leq \frac{1}{p^n}$$

i.e., $v_p(x - a) \geq n$, so $x - a \in p^n \mathbb{Z}_p$ and thus $x \equiv a \pmod{p^n \mathbb{Z}_p}$. So we get the isomorphism

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}$$

□

We will now create the link with previous definitions of the ring $\mathbb{Z}_p$ and the field $\mathbb{Q}_p$. From last proposition we get, $\forall n \geq 1$, a surjective homomorphism

$$\phi_n : \mathbb{Z}_p \longrightarrow \mathbb{Z} / p^n \mathbb{Z}.$$

This family of these homomorphisms induces the homomorphism (from universal property)

$$\mathbb{Z}_p \longrightarrow \varprojlim_n \mathbb{Z} / p^n \mathbb{Z}.$$

Now we can identify both definitions given for $\mathbb{Z}_p$ (and also for $\mathbb{Q}_p$).

(4.18) Proposition

. The homomorphism

$$\Phi : \mathbb{Z}_p \longrightarrow \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

is an isomorphism.

Proof: If $x \in \ker(\Phi)$, then $x \in p^n\mathbb{Z}_p$ for all $n \geq 1$ since $x \in \ker(\phi_n)$, i.e., $|x|_p \leq \frac{1}{p^n}$ for all $n \geq 1$, so that $|x|_p = 0$ and thus $x = 0$. This shows injectivity. Now, any element of $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ is given by a sequence of partial sums

$$s_n = \sum_{i=0}^{n-1} a_i p^i, \quad 0 \leq a_i < p.$$

By (4.6) this sequence is a Cauchy sequence in $\mathbb{Z}_p$, and hence converges to an element

$$x = \sum_{i=0}^{\infty} a_i p^i \in \mathbb{Z}_p.$$

Since

$$x - s_n = \sum_{i=n}^{\infty} a_i p^i \in p^n \mathbb{Z}_p,$$

we have $x \equiv s_n \pmod{p^n}$ for all n , i.e., x is mapped to an element $(s_1, s_2, s_3, \dots)$ of $\varprojlim \mathbb{Z}/p^n \mathbb{Z}$ which is defined by the given sequence $(s_n)_{n \in \mathbb{N}}$. This proves surjectivity. □

(5.1) Definition .

A valuation (multiplicative valuation or absolute value) on a field K is a function

$$|\cdot| : K \rightarrow \mathbb{R}$$

satisfying the properties, $\forall x, y \in K$:

1. $|x| = 0 \Leftrightarrow x = 0$,
2. $|x| \geq 0$,
3. $|xy| = |x||y|$,
4. $|x + y| \leq |x| + |y|$. (Triangle inequality)

(5.2) Definition .

Two valuations on K are equivalent if they define the same topology on K .

(5.3) Theorem

. Let $|\cdot|_1$ and $|\cdot|_2$ be two valuations on K . Then, they are equivalent iff, $\exists$ a real $s \geq 0$ s.t

$$|x|_1 = |x|_2^s, \forall x \in K.$$

Proof: If we have $|x|_1 = |x|_2^s, s > 0$, clearly the two valuations are equivalent since for $\epsilon > 0$,

$$B_{d_1}(x, \epsilon^s) \subseteq B_{d_2}(x, \epsilon) \text{ and } B_{d_2}(x, \epsilon^{1/s}) \subseteq B_{d_1}(x, \epsilon)$$

For converse, if $|\cdot|_1$ and $|\cdot|_2$ are equivalent, a series converging to 0 w.r.t $|\cdot|_1$ will converge to 0 with respect to $|\cdot|_2$ and vice versa. Also, for all K field and $|\cdot|$ on K , the inequality $|x| < 1$ is equivalent to saying that the sequence $\{x^n\}_{n \in \mathbb{N}}$ converges to 0. Therefore, we have

$|\cdot|_1$ and $|\cdot|_2$ are equivalent $\implies$

$$(x^n \rightarrow 0 \text{ w.r.t } |\cdot|_1 \iff x^n \rightarrow 0 \text{ w.r.t } |\cdot|_2) \iff (|x|_1 < 1 \iff |x|_2 < 1).$$

Take $y \in K$ an element s.t $|y|_1 > 1$ and let $x \in K, x \neq 0$. Then there exists $\alpha \in \mathbb{R}$ such that $|x|_1 = |y|_1^\alpha$. Let $\left\{ \frac{m_i}{n_i} \right\}_{i \in \mathbb{N}}$ be a sequence of rational numbers ($m_i \in \mathbb{Z}, n_i \in \mathbb{N}$) that converges to α from above. Then $|x|_1 = |y|_1^\alpha < |y|_1^{\frac{m_i}{n_i}}$ as a^x for $a > 1$ is increasing function and hence

$$\left| \frac{x^{n_i}}{y^{m_i}} \right|_1 < 1 \Rightarrow \left| \frac{x^{n_i}}{y^{m_i}} \right|_2 < 1$$

Hence, $|x|_2 < |y|_2^{\frac{m_i}{n_i}} \Rightarrow |x|_2 \leq |y|_2^\alpha$. Now, take a sequence $\left\{ \frac{m_i}{n_i} \right\}_{i \in \mathbb{N}}$ that converges to α from below. It gives $|x|_2 \geq |y|_2^\alpha$ and therefore $|x|_2 = |y|_2^\alpha$. So, for all $0 \neq x \in K$, we have

$$\frac{\log |x|_1}{\log |x|_2} = \frac{\log |y|_1}{\log |y|_2} =: s$$

and thus $|x|_1 = |x|_2^s$. Also, $|y|_1 > 1 \Rightarrow |y|_2 > 1$ from equivalence and so $s > 0$.

(5.4) Definition

. A valuation is **non-Archimedean** if $|n|$ is bounded, $\forall n \in \mathbb{N}$. Otherwise, called **Archimedean**

(5.5) Proposition .

The valuation $|\cdot|$ is nonarchimedean iff it satisfies the given strong triangle inequality

$$|x + y| \leq \max\{|x|, |y|\}.$$

Proof: If this inequality holds, then we have

$$|n| = |1 + \cdots + 1| \leq 1.$$

For converse, let $|n| \leq N \forall n \in \mathbb{N}$. Let $x, y \in K$ and suppose $|x| \geq |y|$. Then $|x|^\nu |y|^{n-\nu} \leq |x|^n$ for $\nu \geq 0$ and we get

$$|x + y|^n \leq \sum_{\nu=0}^n \left| \binom{n}{\nu} \right| |x|^\nu |y|^{n-\nu} \leq N(n+1) |x|^n,$$

thus

$$|x + y| \leq N^{1/n}(1 + n)^{1/n}|x| = N^{1/n}(1 + n)^{1/n} \max\{|x|, |y|\},$$

and hence $|x + y| \leq \max\{|x|, |y|\}$ by letting $n \rightarrow \infty$. □

(5.6) Ostrowski's Theorem .

Every valuation on $\mathbb{Q}$ is equivalent to the valuations $|\cdot|_p$ or $|\cdot|_\infty$.

Proof. Let $\|\cdot\|$ be a non-Archimedean valuation on $\mathbb{Q}$. Then,

$$\|n\| = \|1 + \cdots + 1\| \leq \max\{\|1\|, \dots, \|1\|\} = 1$$

There also exists a prime $p > 1$ such that $\|p\| < 1$. If not, then all $\|p\| = 1$ and so by unique prime factorization $\|\cdot\|$ is the trivial valuation. Now, consider the set

$$A = \{a \in \mathbb{Z} \mid \|a\| < 1\}.$$

Here, A is an ideal of $\mathbb{Z}$ (For, $a, b \in A$ and $r \in \mathbb{Q}$, $\|a + b\| \leq \max\{\|a\|, \|b\|\} < 1$, and $\|ra\| = \|r\|\|a\| < 1$). Also $p\mathbb{Z} \subseteq A \neq \mathbb{Z}$ ($\|1\| = 1$ and $1 \in \mathbb{Z}$). Since $p\mathbb{Z}$ is a maximal ideal, we get that $A = p\mathbb{Z}$.

Take $a \in \mathbb{Z}$, $a = p^m b$ with $p \nmid b$. Then $p \nmid b \implies b \notin A$ and thus $\|b\| = 1$. This provides

$$\|a\| = \|p^m b\| = \|p\|^m = |a|_p^s,$$

with $s := -\frac{\log \|p\|}{\log p}$. Hence, $\|\cdot\|$ is equivalent to $|\cdot|_p$. Now assume that $\|\cdot\|$ is Archimedean.

Claim: For all integer $m, n > 1$, we have

$$\|m\|^{\frac{1}{\log m}} = \|n\|^{\frac{1}{\log n}}.$$

Proof. we write

$$m = a_0 + a_1 n + \cdots + a_r n^r$$

with $r, a_i \in \mathbb{N}$, $0 \leq a_i < n$, $0 \leq i \leq r$ and $n^r \leq m$, $\implies r \leq \frac{\log m}{\log n}$. Also,

$$\|a_i\| \leq \|1 + \cdots + 1\| \leq a_i \|1\| \leq n$$

gives us that

$$\|m\| \leq \sum_{i=0}^r \|a_i\| \|n\|^i \leq \sum_{i=0}^r \|a_i\| \|n\|^r \leq \left(1 + \frac{\log m}{\log n}\right) n \|n\|^{\frac{\log m}{\log n}}$$

Replace m by m^k , $k \in \mathbb{N}$, and take the k -th root, then we obtain that

$$\|m\| = \sqrt[k]{\|m^k\|} \leq \sqrt[k]{\left(1 + \frac{k \log m}{\log n}\right) n \|n\|^{\frac{k \log m}{\log n}}} = \|n\|^{\frac{\log m}{\log n}} \sqrt[k]{\left(1 + \frac{k \log m}{\log n}\right) n}$$

So when $k \rightarrow \infty$ we have

$$\|m\| \leq \|n\|^{\frac{\log m}{\log n}} \text{ and so } \|m\|^{\frac{1}{\log m}} \leq \|n\|^{\frac{1}{\log n}}.$$

Reversing the roles of m and n we get the second inequality. Since

$\|n\|^{\frac{1}{\log n}} > 0$, $\exists s \in \mathbb{R}$ s.t. $e^s = \|n\|^{\frac{1}{\log n}} \implies \|n\| = e^{s \log n}$. Therefore, $\forall x \in \mathbb{Q}, x > 0$, we have

$$\|x\| = e^{s \log x} = x^s = |x|^s$$

As $\|x\| = \|-x\|$, we have that $\|\cdot\|$ is equivalent to $|\cdot|_\infty$. $\square$

References

- Neukirch Algebraic Number Theory
- Commmutative Algebra by Atiyah